$SAAFL^*$: Secure Aggregation for Label-Aware Federated Learning

* $SAAFL$ means “simple” in $Urdu$

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21/05/2025

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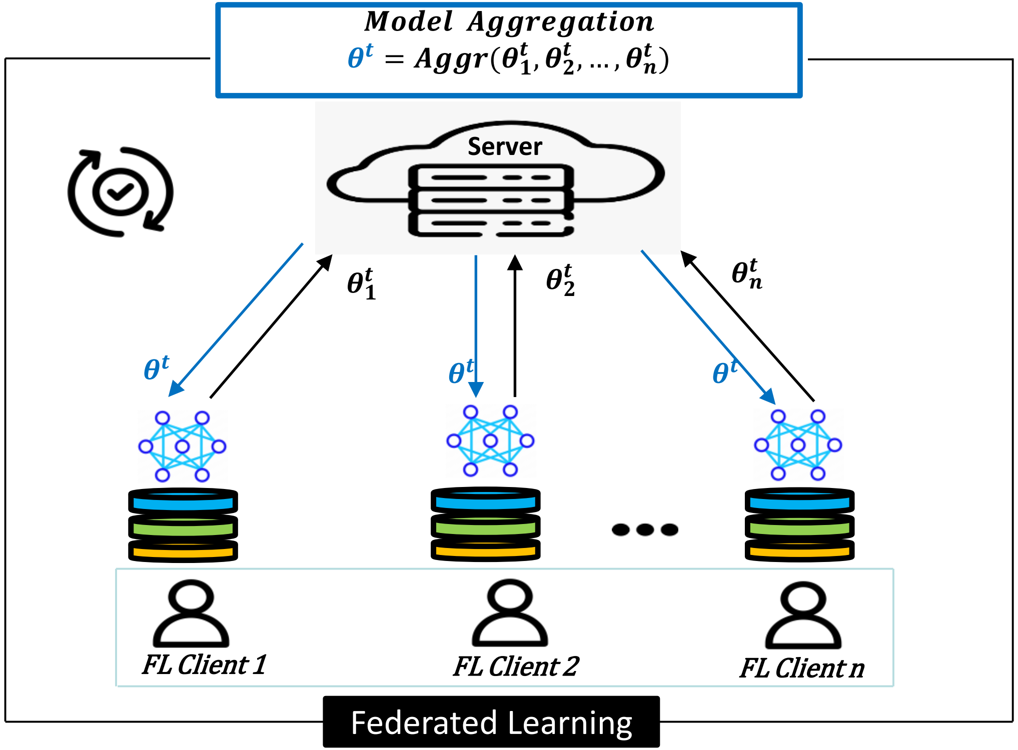
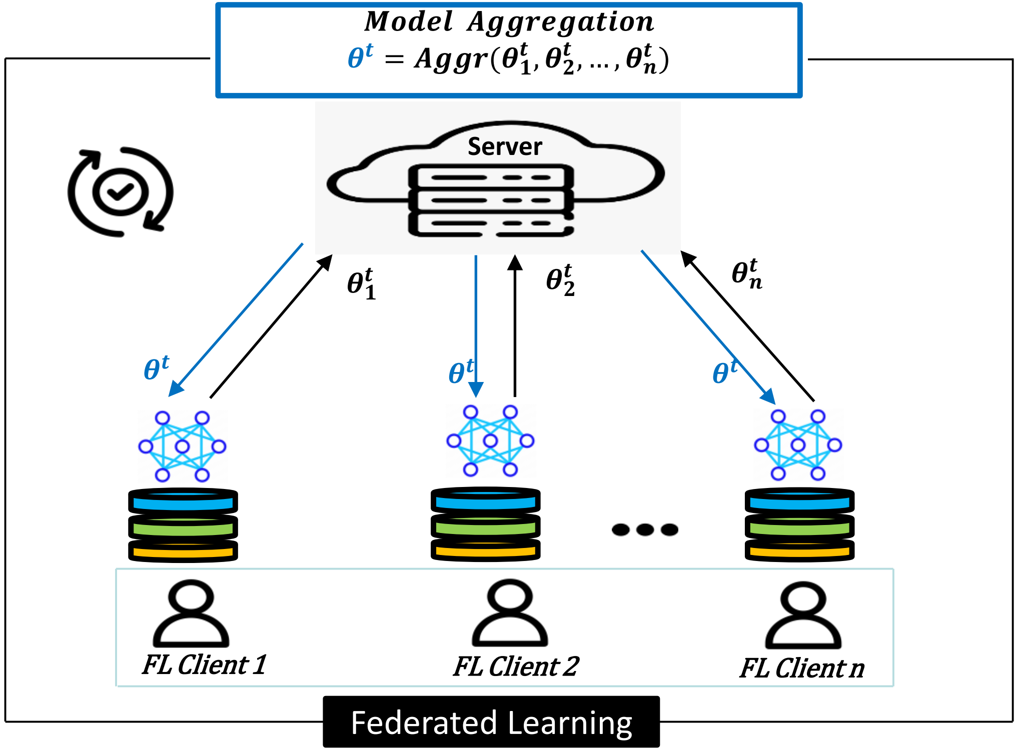
Federated learning (FL)

- Collaborative learning
- Each client trains locally
- Global model = sum of local gradients
- Iterates until optimal accuracy



FL aggregation techniques

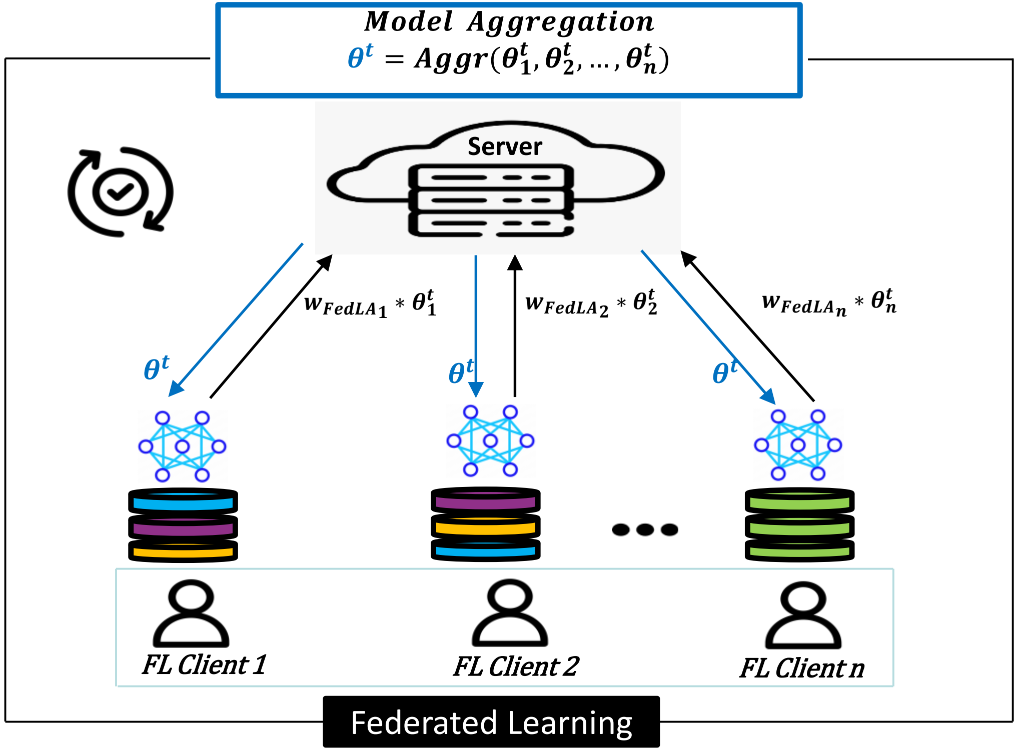
- **Federated Average (FedAvg)** [1]
  - Computes average over gradients
  - $\boldsymbol{\theta^t} = \frac{\sum_{i=1}^{n} \theta_i^t}{n}$ t-th FL round
  - Works well for only **IID** datasets
    - IID = independent and identical distribution



[1] McMahan, B., Moore, E., Ramage, D., Hampson, S., Aguera y Arcas, B.: Communication efficient learning of deep networks from decentralized data. In: International Conference on Artificial Intelligence and Statistics (AISTATS) (2017)

FL aggregation techniques

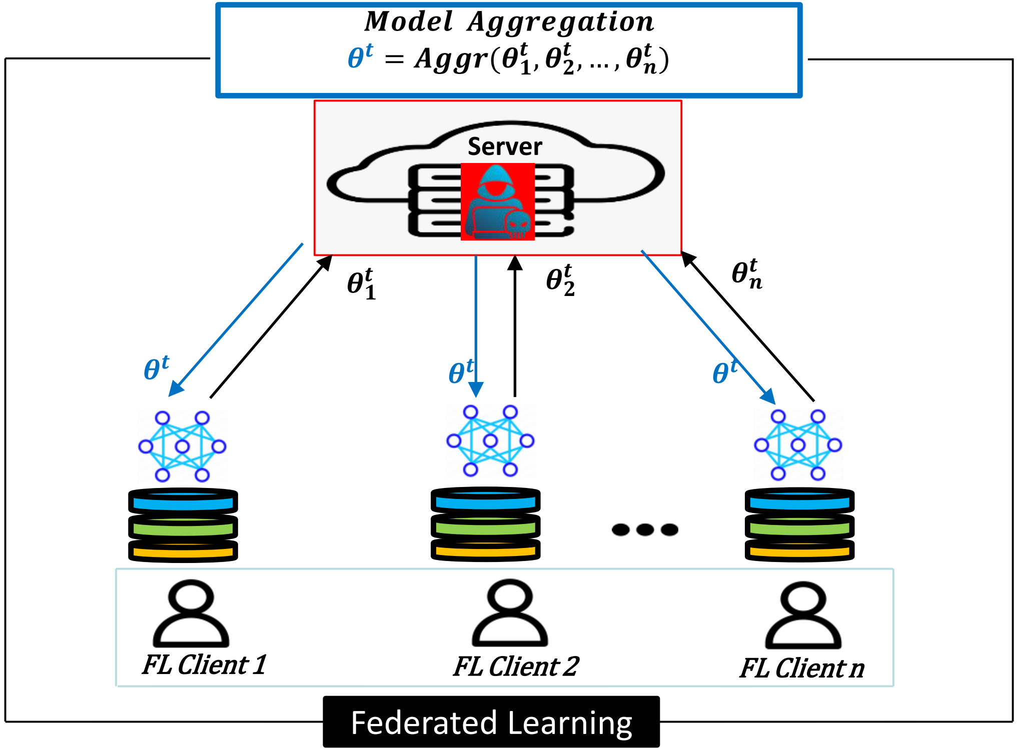
- **Federated Average (FedAvg)** [1]
  - Computes average over gradients
  - $\boldsymbol{\theta^t} = \frac{\sum_{i=1}^{n} \theta_i^t}{n}$ t-th FL round
  - Works well for only **IID** datasets
    - IID = independent and identical distribution
- **Federated Label-Aware (FedLA)** [2]
  - Computes average over weighted gradients
  - $\boldsymbol{\theta^t} = \frac{\sum_{i=1}^{n} w_{FedLA_i} * \theta_i^t}{n}$
  - Weights depend clients label distributions
  - Designed to handle the **non-IID** data



[2] Khalil, A., Wainakh, A., Zimmer, E., Parra-Arnau, J., Anta, A.F., Meuser, T.,Steinmetz, R.: Label-aware aggregation for improved federated learning. In: Inter-national Conference on Fog and Mobile Edge Computing (FMEC) (2023)

Privacy risks in FL

- **Privacy Attacks**
  - **Membership inference attack [3]**
    - Infers data point in dataset
  - **Data property inference attack [4]**
    - Infers sensitive attributes in dataset



[3] R. Shokri, M. Stronati, C. Song, V. Shmatikov, Membership inference attacks against machine learning models, in: 2017 IEEE Symposi Security and Privacy, 2017, pp. 3–18

[4] K. Ganju, Q. Wang, W. Yang, C. A. Gunter, N. Borisov, Property inference attacks on fully connected neural networks using permutation invariant representations, in: 2018 ACMSIGSAC Conference on Computer and Communications Security, 2018, pp. 619–633.

Secure aggregation (SA)

- Aggregates without revealing inputs
- **SA** can be achieved using:
  - **Homomorphic encryption**
  - Masking
  - Multiparty computation

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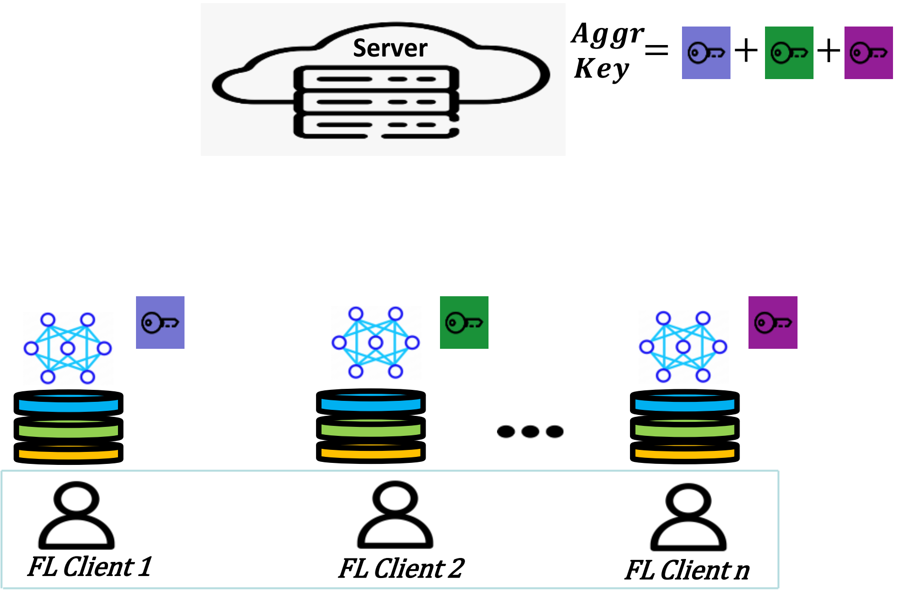
  - Key setup phase



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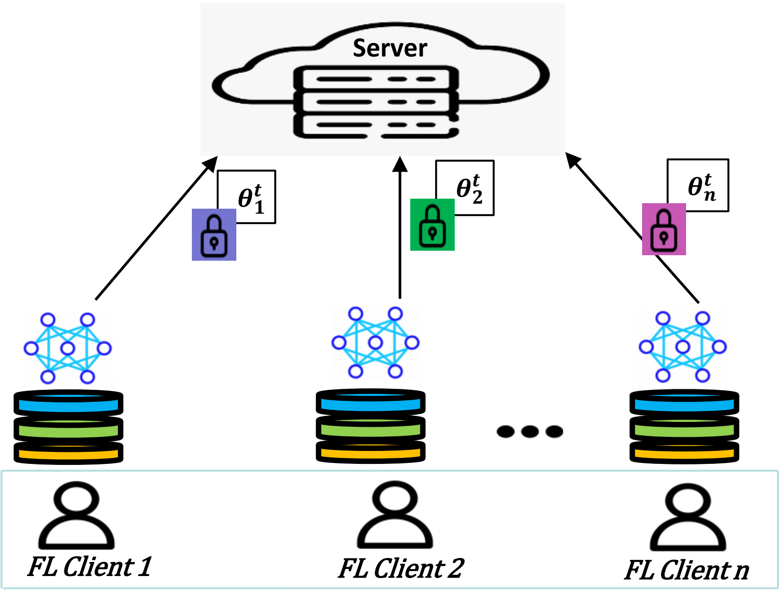
  - Protection phase



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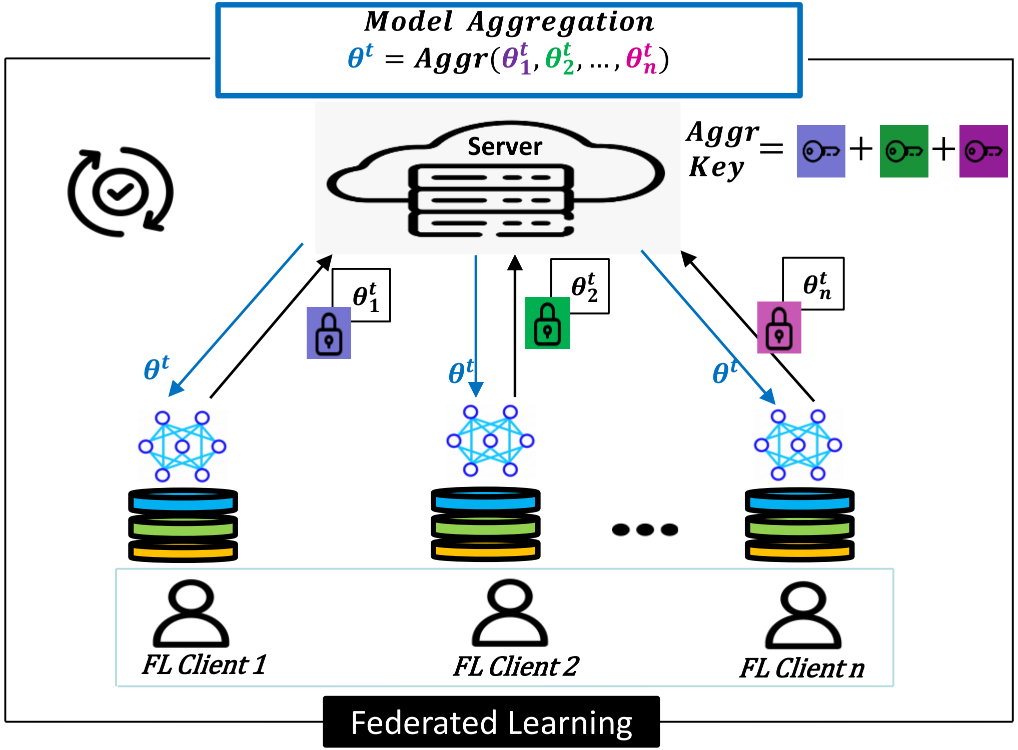

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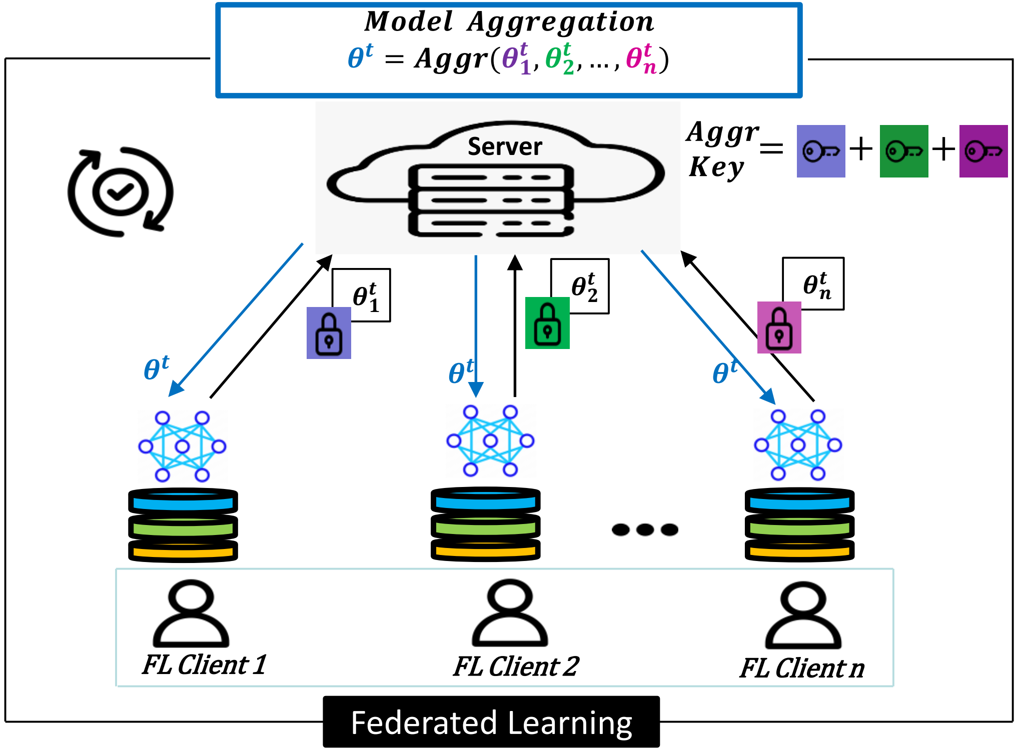
- **SA** consists of the following steps:
  - Key setup phase
  - Protection phase
  - Aggregation phase
- In literature, **SA** mostly targets **FedAvg** [5]
- **FedAvg-based SA** fails for **FedLA** due to:
  - Weights dependence
  - Needs two **SA** for weights and gradients



[5] Mansouri, M. and Önen, M. and Ben Jaballah, W. and Conti, M.: Sok: Secure aggregation based on cryptographic schemes for federated learning. In: Proceedings on Privacy Enhancing Technologies (PoPETS) (2023)

Our goal

How to design SA for FedLA?

Outline

- Privacy-preserving FedLA: Challenges
  - Data privacy
  - Client dropouts
  - Client selection
- SAAFL: Our solution
  - Secure aggregation for data privacy
  - Fault-tolerant FedLA
  - Handling client selection
- Evaluation
  - Model accuracy
  - Computation and communication cost

Privacy-preserving FedLA: Challenges

- **Weight Calculation**
  1. **Client weight per label:**

     $s_l = \sum_{i=1}^{n} S_{i,l}$; $S_{i,l}$=no. of samples of label l of client i

     $$w_{i,l} = \frac{S_{i,l}}{s_l}$$

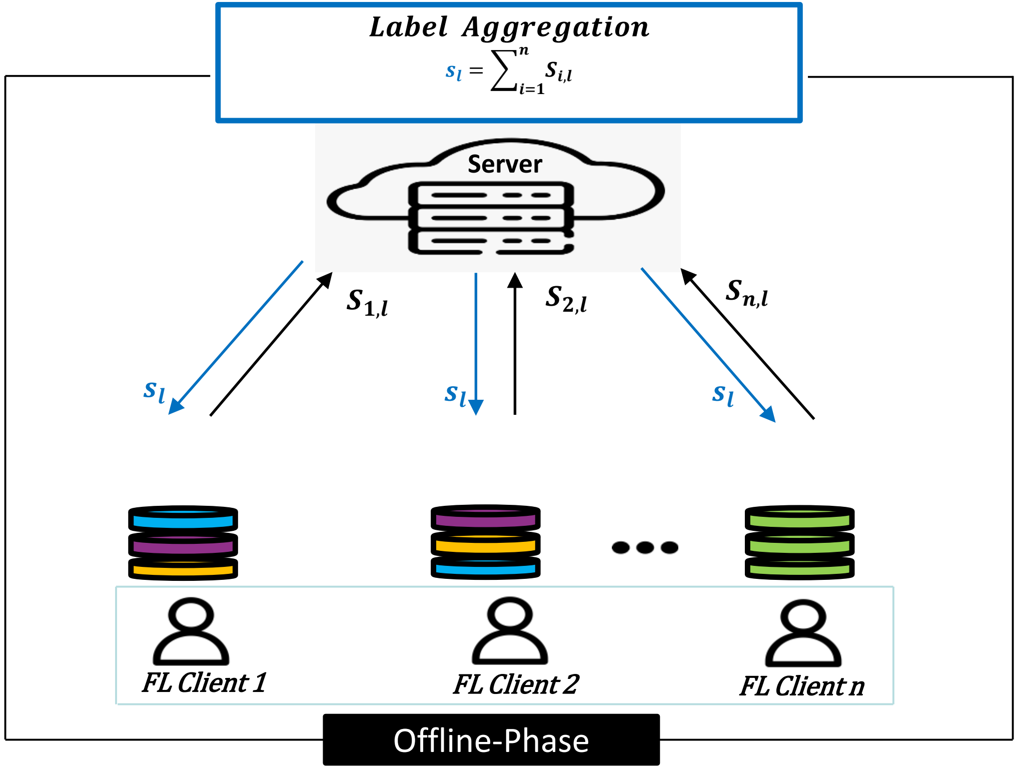
  2. **Client weight w.r.t all labels:**

     $$w_i = \sum_{l=1}^{k} w_{i,l}$$

  3. **FedLA weight:**

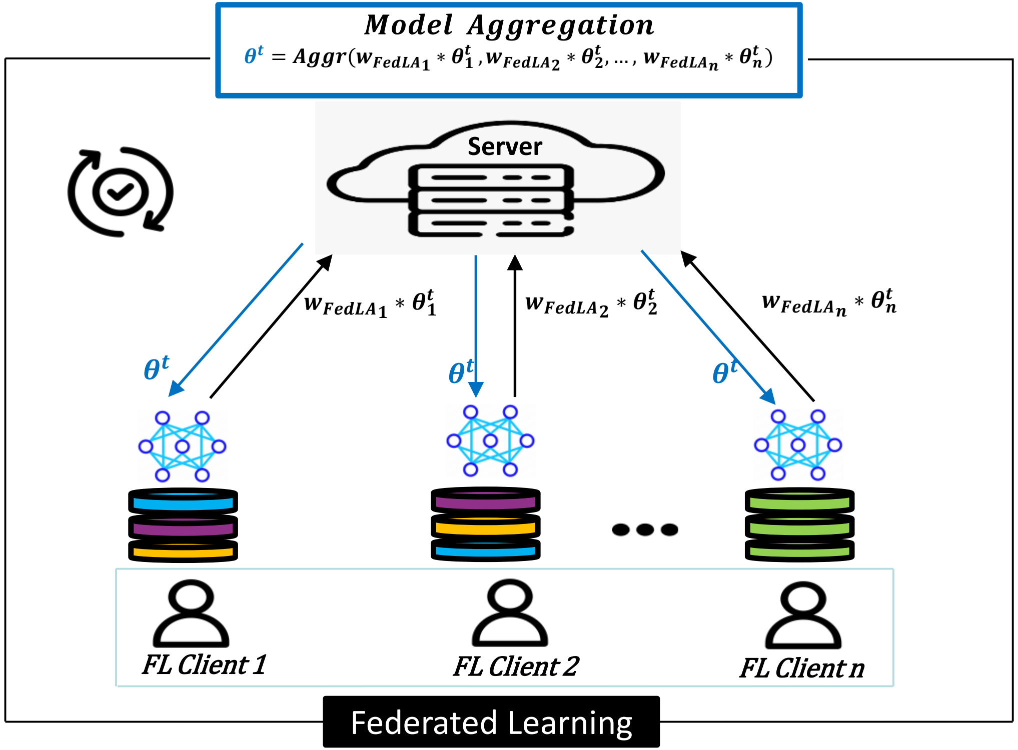
     $$w_{FedLA,i} = \frac{w_i}{\sum_{i=1}^{n} w_i}$$

     $\sum_{i=1}^{n} w_i$ = k = Total number of labels of dataset



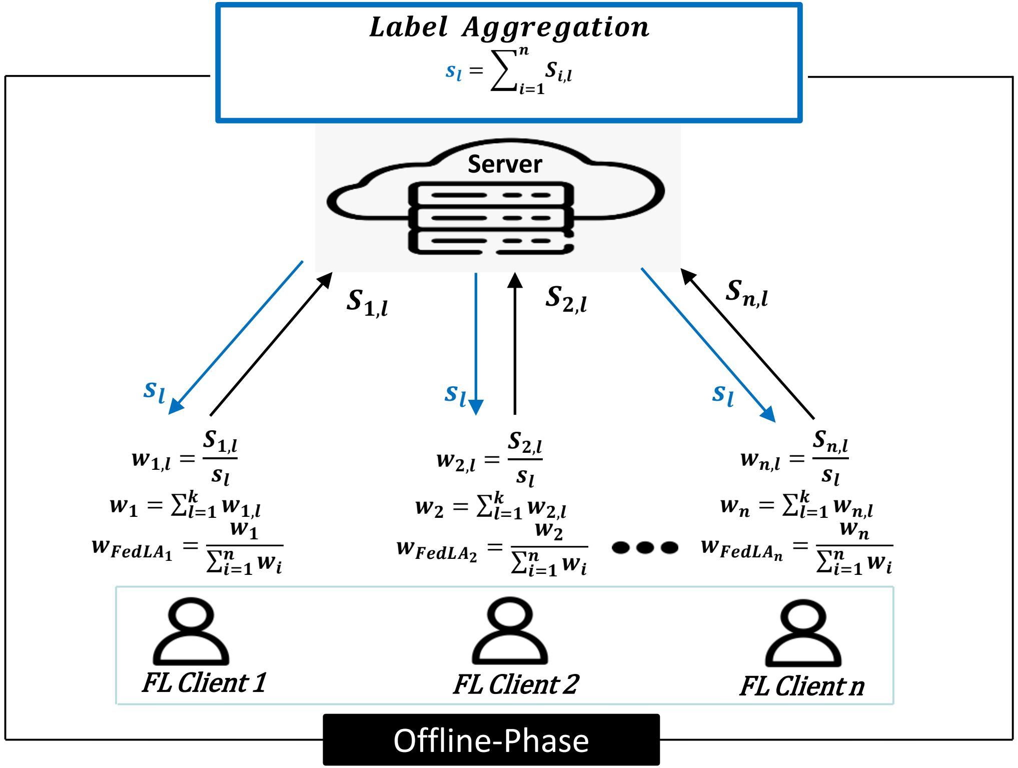
Privacy-preserving FedLA: Challenges

- **Model privacy**
  - Clients send weighted gradients to server
  - Requires **SA** for **model aggregation**



Privacy-preserving FedLA: Challenges

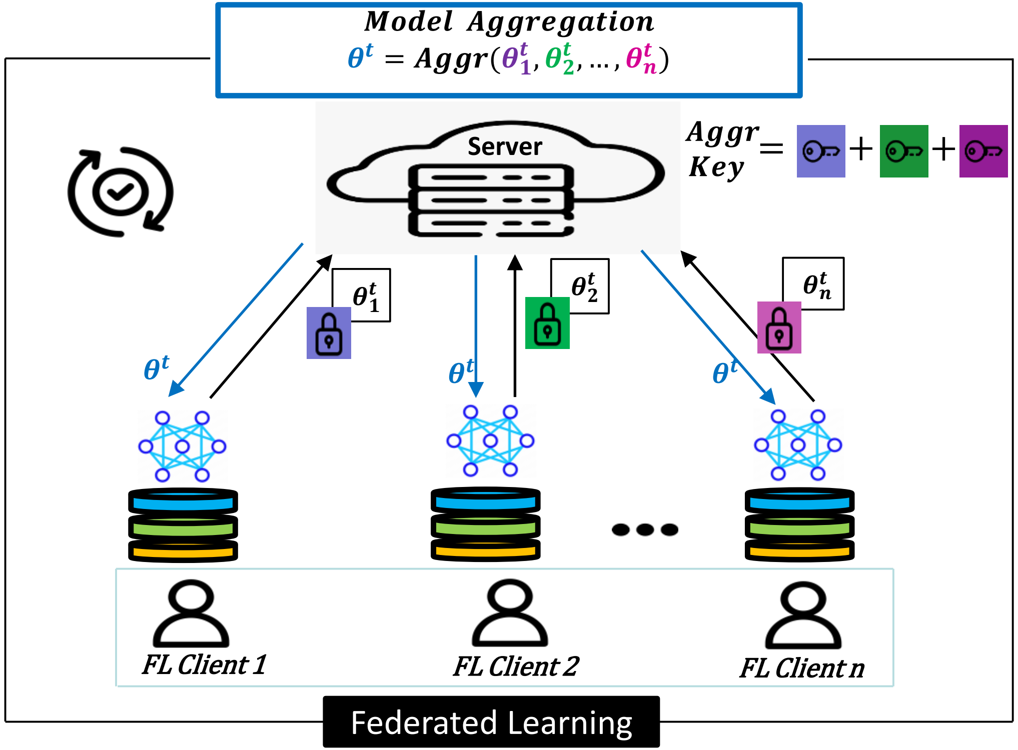
- **Label privacy**
  - Since these weights (w_{FedLA}) are dependent
  - Exposing label weights breaches privacy [6]
    - Label count infers about data in graph neural network
  - Requires another **SA** for w_{FedLA} computation



[6] Zari, O., Xu, C., Parra-Arnau, J., Ünsal, A., Önen, M.: Link Inference Attacks inVertical Federated Graph Learning. In: 40th Annual Computer Security Applica-tions Conference (ACSAC) (2024)

Privacy-preserving FedLA: Challenges

- **Client dropouts**
  - Aggregation key depends on all clients keys



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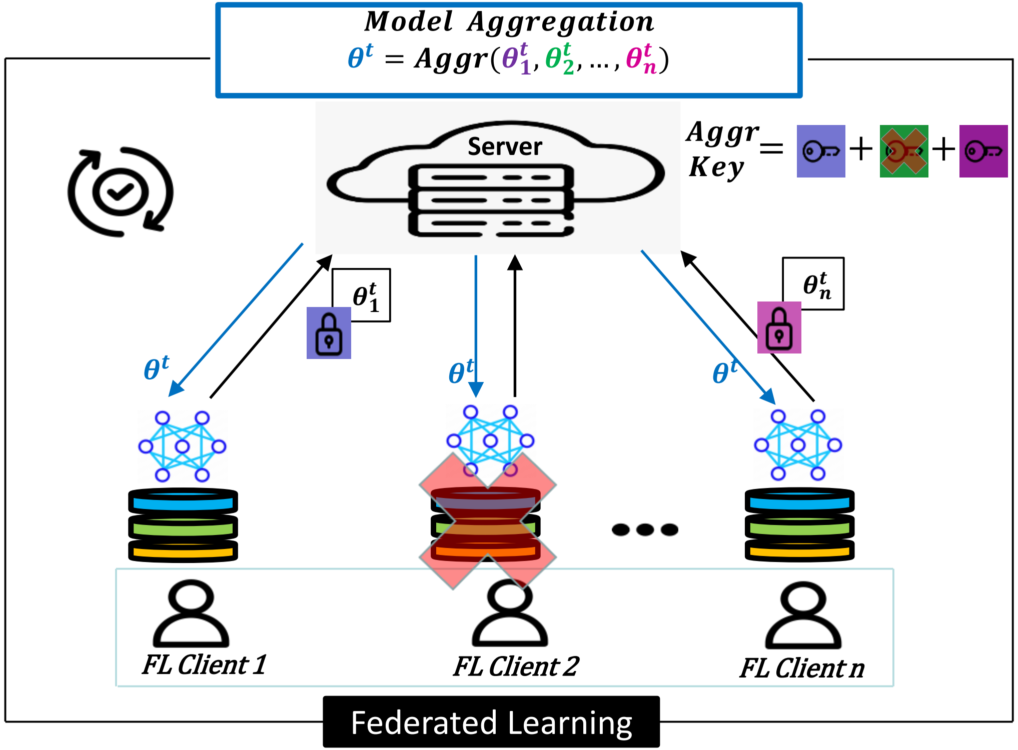

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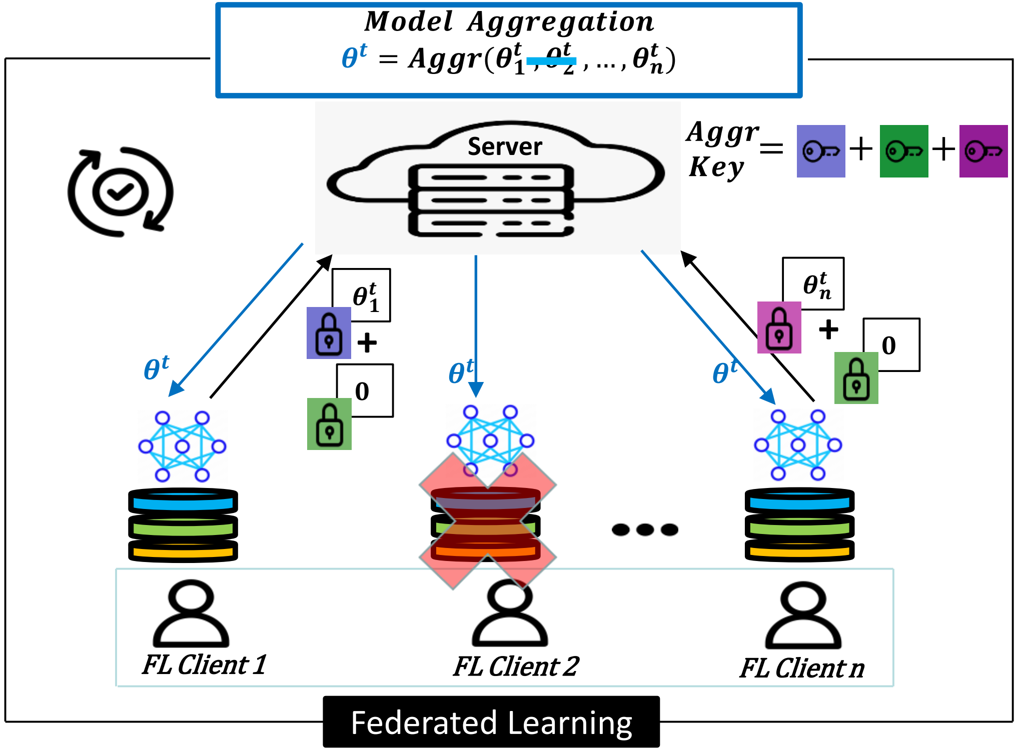
  - Previous works use **zeros** for dropouts [7]



[7] Mansouri, M., Önen, M., Ben Jaballah, W.: Learning from Failures: Secure and Fault-Tolerant Aggregation for Federated Learning. In: 38th Annual Computer Security Applications Conference (ACSAC) (2022)

Privacy-preserving FedLA: Challenges

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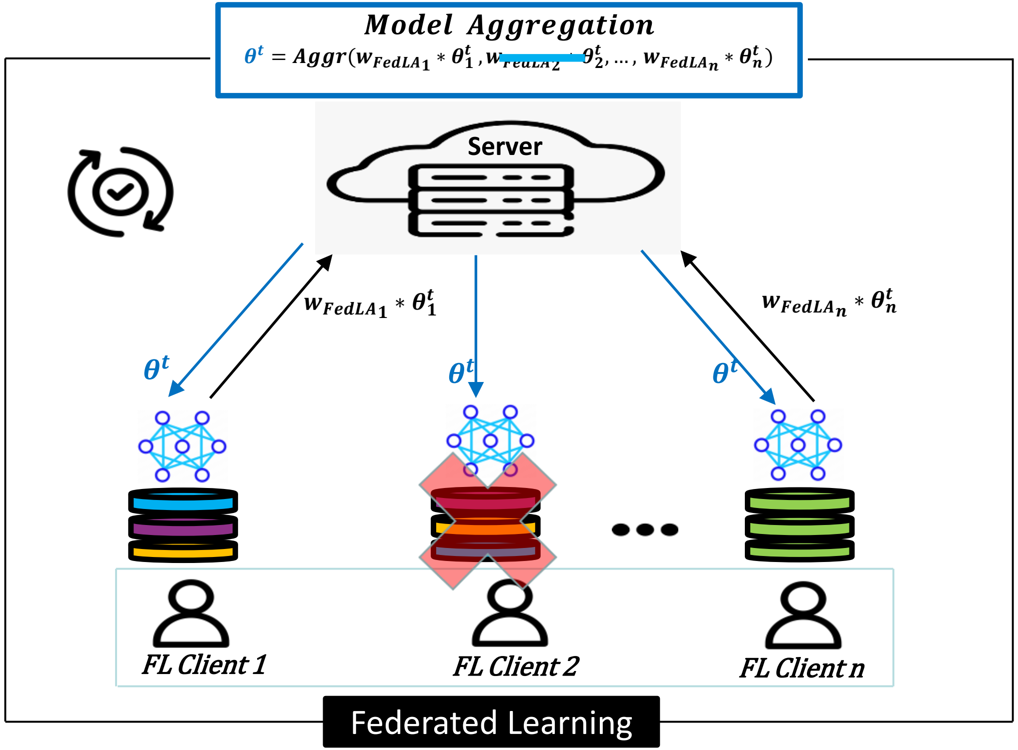

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- **Client dropouts**
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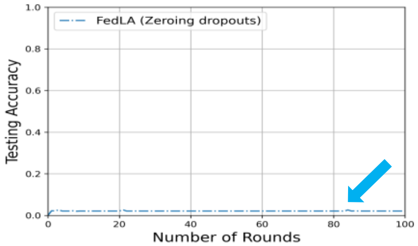
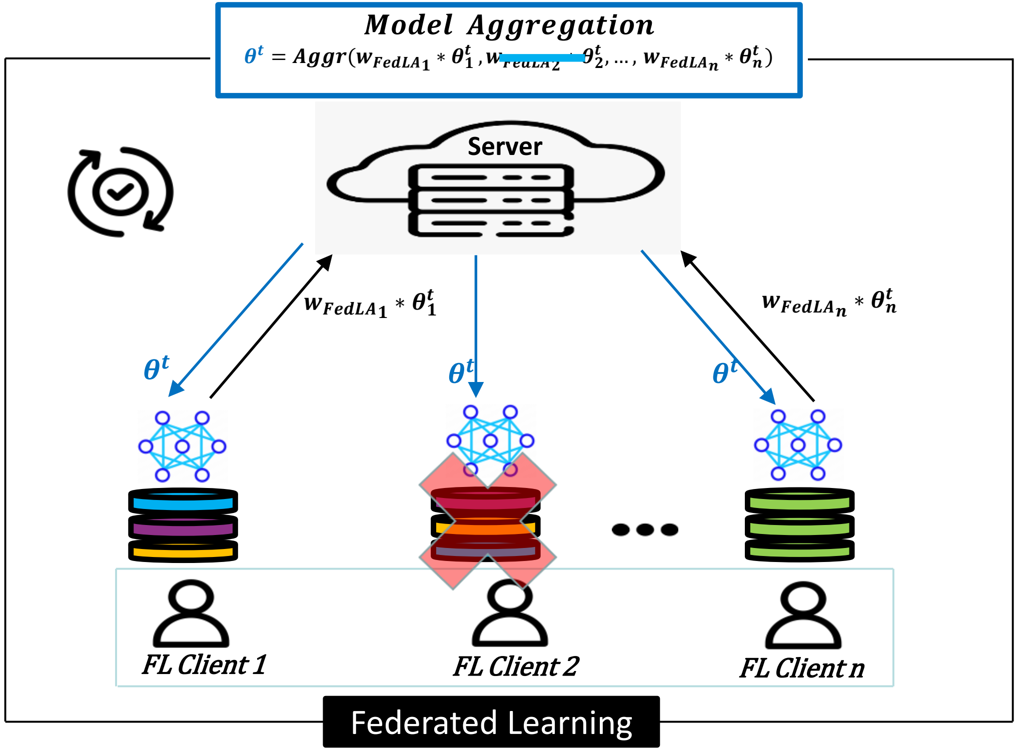
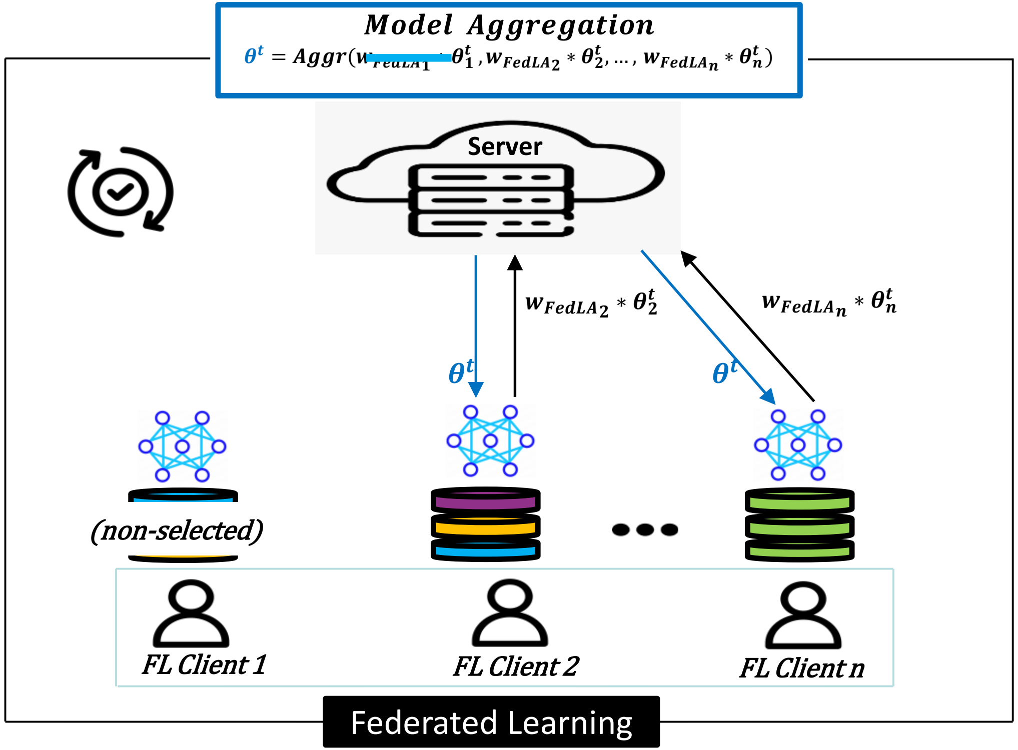
  - Client dropout impact aggregation
  - Previous works use **zeros** for dropouts [7]
  - Since weights (w_{FedLA}) are dependent
  - Zeroing dropouts reduces performance in **FedLA**
  - Requires dedicated **fault-tolerant SA**





Privacy-preserving FedLA: Challenges

- **Client selection in FedLA**
  - In **FedLA**, client selection occurs each round
  - Since, weights (w_{FedLA}) are dependent
  - This leads to dropout problem
  - Needs **SA** tolerant to non-selected clients



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SAAFL: Our solution

- **Weight Calculation (Recap)**

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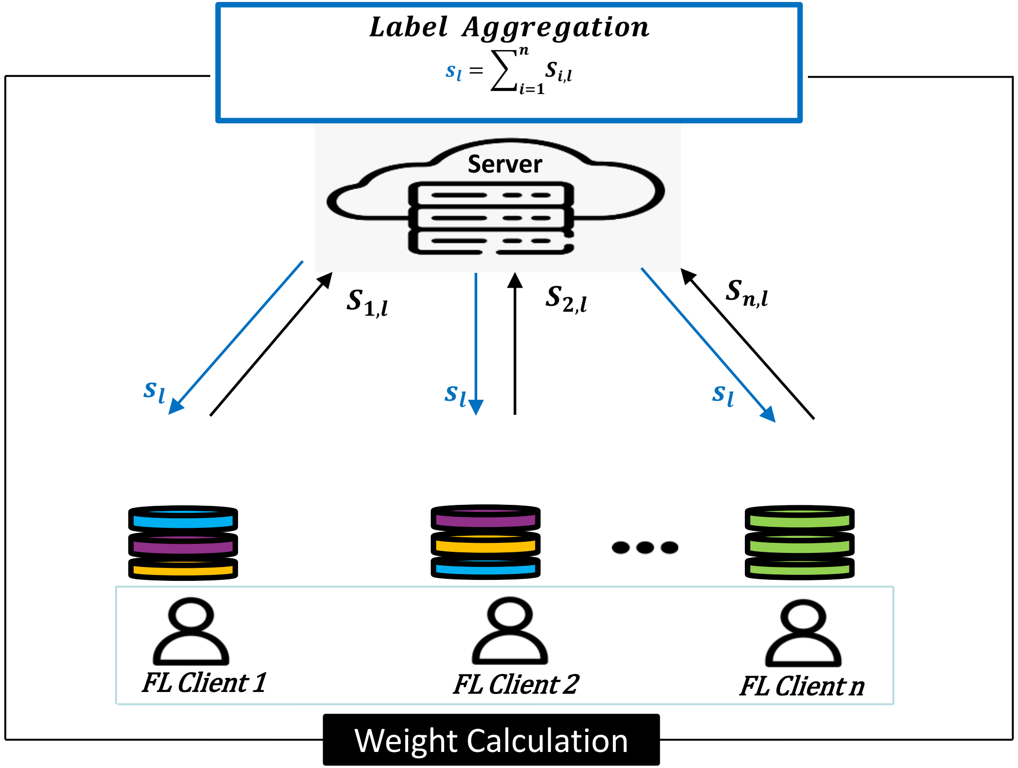

Secure aggregation for data privacy

- **First SA for labels**
  - Client i encrypts label l sample count ($S_{i,l}$)
  - Server performs **SA** over $S_{i,l}$, returns s_l
  - Client computes w_{FedLA} as following:
    1. $w_{i,l} = \frac{S_{i,l}}{s_l}$
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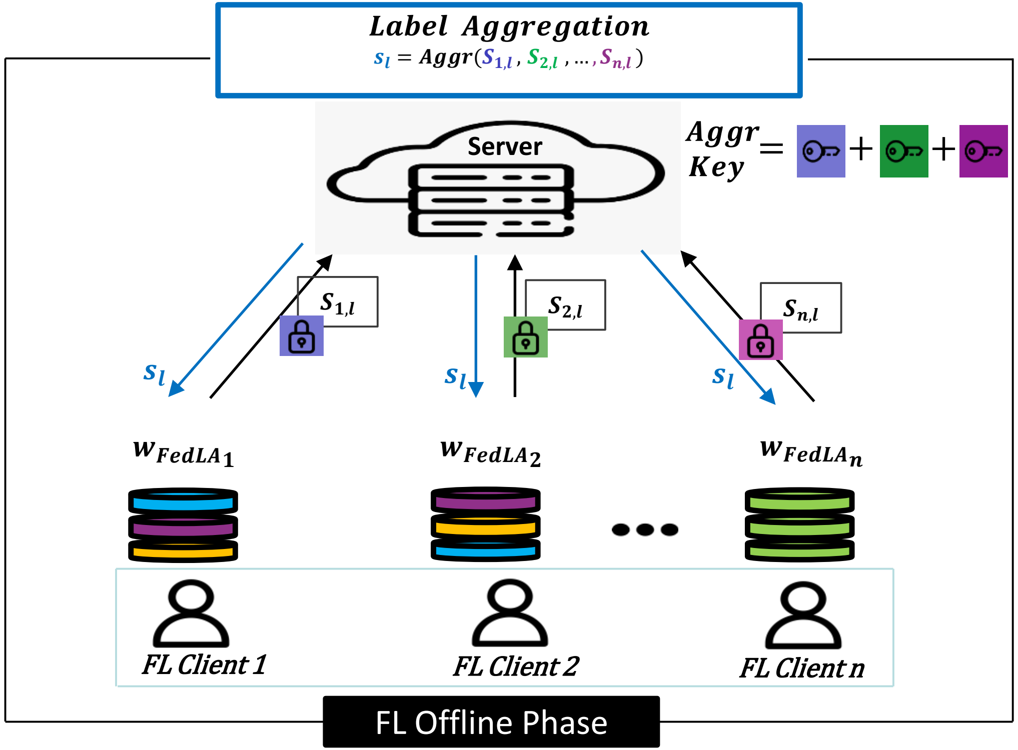

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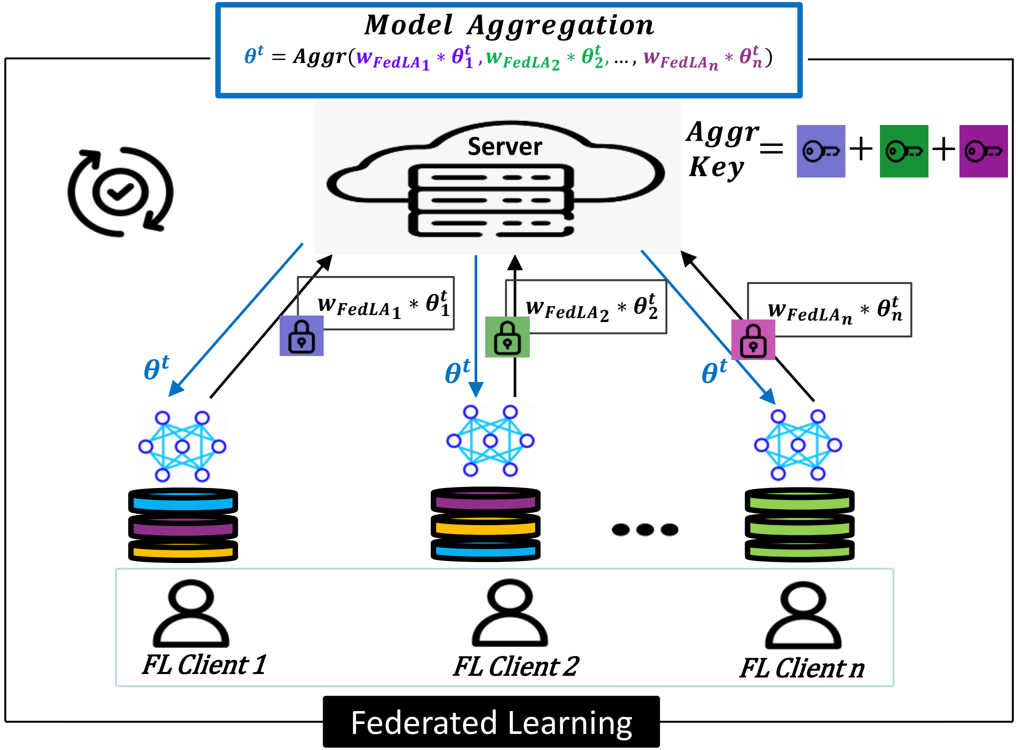
    3. $w_{FedLA,i} = \frac{w_i}{\sum_{i=1}^{n} w_i}$
- **Second SA for model**
  - Client encrypts weighted gradients
  - Server performs simple **SA** over $w_{FedLA_i} * \theta_i^t$

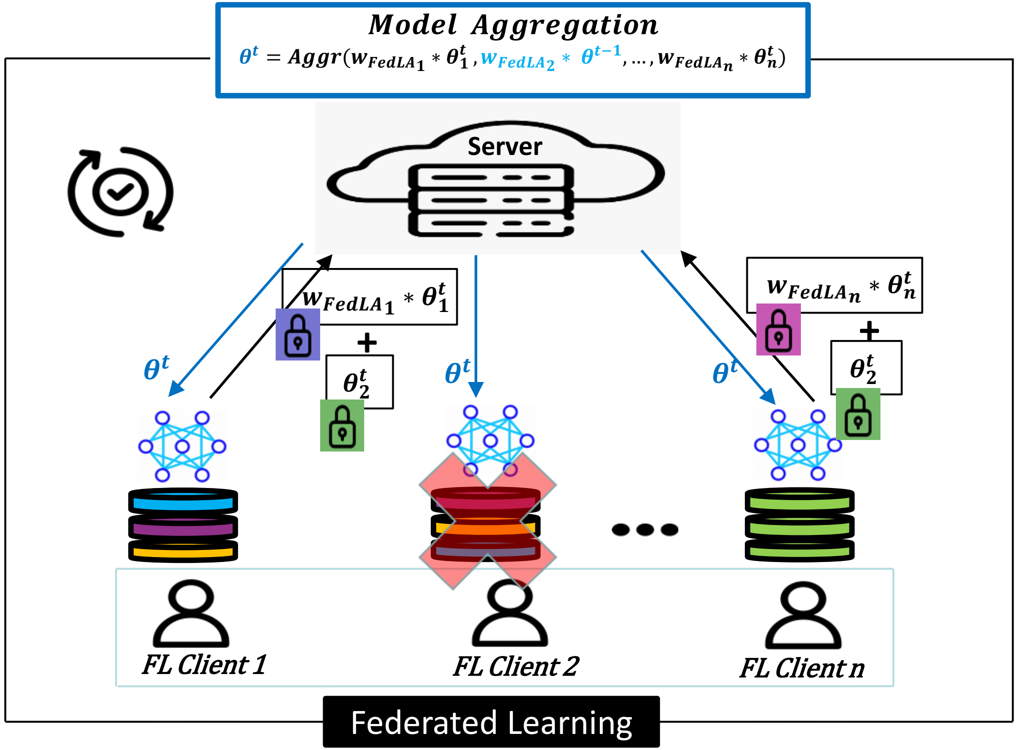


Fault-tolerant FedLA

- **Handling dropouts**
  - Online clients compute inputs for dropouts

$$\theta_2^t = w_{FedLA_2} * \theta^{t-1} = \left(1 - \sum_{i \in U_{on}} w_{FedLA_i}\right)\theta^{t-1}$$

    - $\sum_{i \in U_{on}} w_{fedLA_i}$ = Sum of w_{FedLA} of online clients
    - θ^{t-1} = Previous round model
    - U_{on}= Set of online clients



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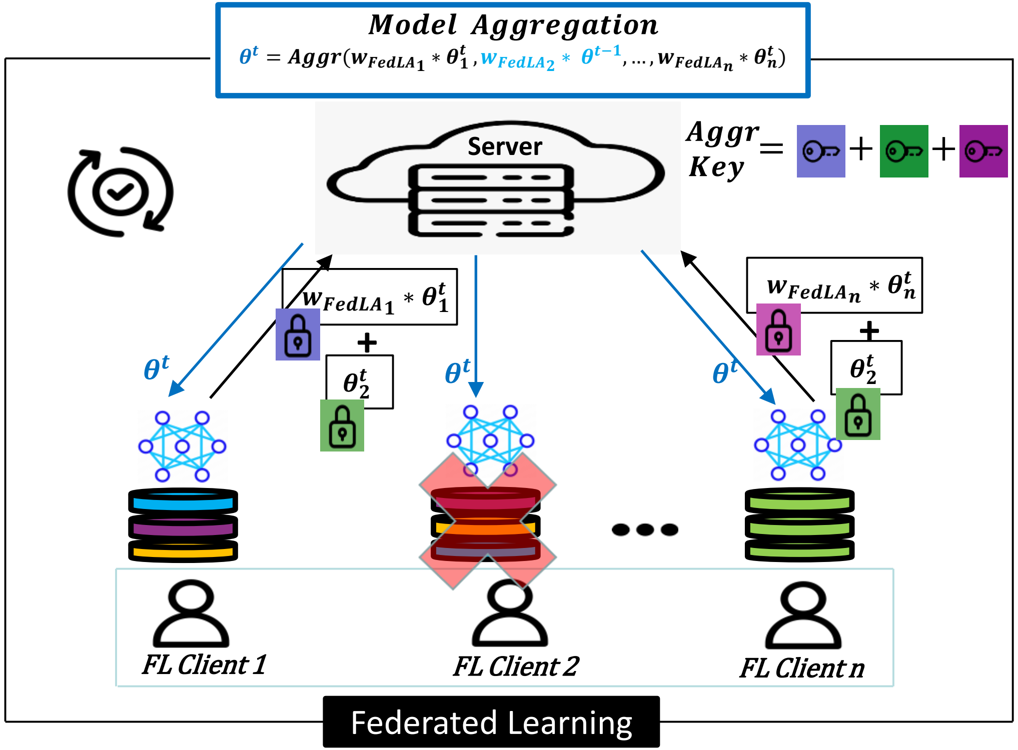
  - Final **SA** including dropout clients inputs:

$$\theta^t = \underbrace{\sum_{i \in U_{on}} w_{fedLA_i} \theta_i^t}_{\text{Online clients inputs}} + \underbrace{(1 - \sum_{i \in U_{on}} w_{fedLA_i})\theta^{t-1}}_{\text{Offline clients inputs}}$$

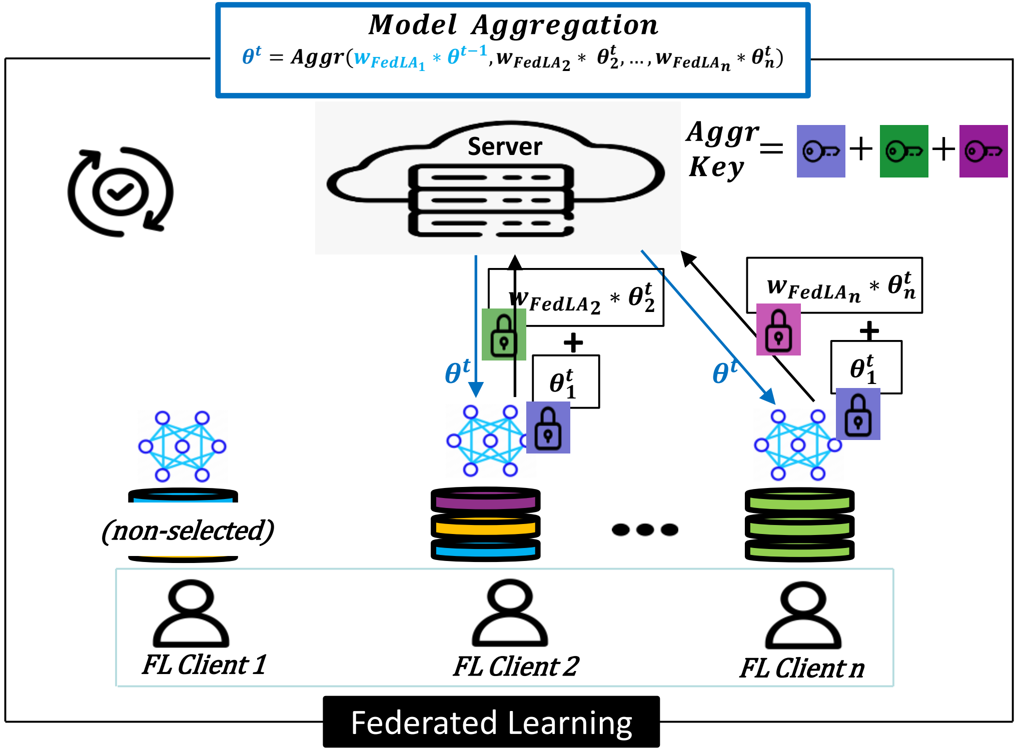


Handling client selection

- **Handling non-selected clients**
  - As non-selected clients treated as dropouts
  - Same dropout handling is applied
  - Selected clients compute inputs for non-selected

$$\theta_1^t = w_{FedLA_1} * \theta^{t-1} = (1 - \sum_{i \in U_{selected}} w_{FedLA_i})\theta^{t-1}$$

  - $U_{selected}$ = Set of selected clients



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Evaluation

- **Evaluation metrics**
  - Model accuracy over
    - FedAvg
    - FedLA with zeroing dropouts
    - SAAFL
  - Computation and communication cost
- **Experimental setup**

Dataset	Number of Clients (n)	Dropout rate (f)	*Non-IID distribution*
EMNIST (47×2400 samples)	{10,100}	$\{0.1, 0.2, 0.3\}$	$\boldsymbol{k_r = \{0.7, 0.8, 0.9\}}$ $\boldsymbol{unique_c = 1}$ $\boldsymbol{noniid_s = 0.7}$

k_r = selection ratio
$noniid_s$ = non-IID client ratio
$unique_c$ = Unique labels per non-IID client

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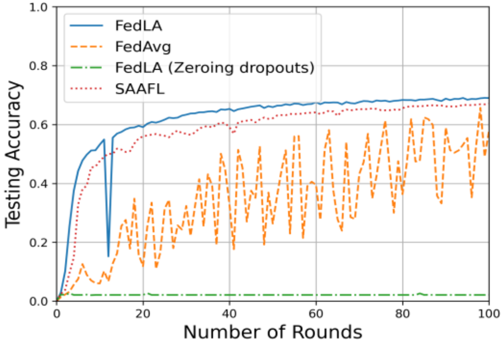
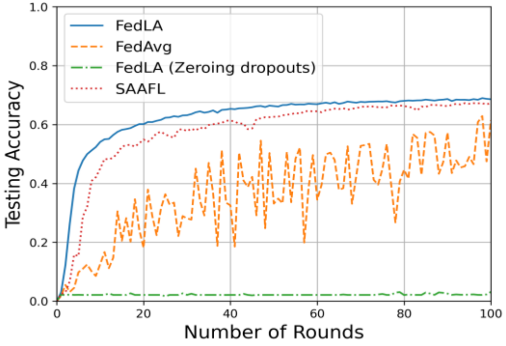
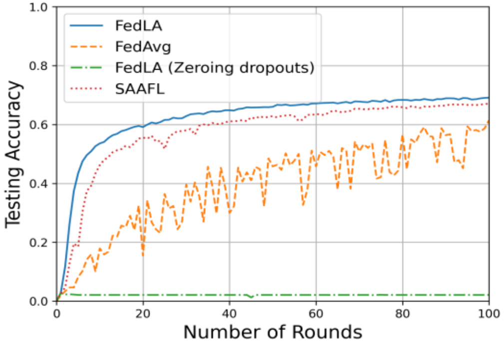
Accuracy over EMNIST dataset



$k_r = 0.7, f = 0.1$

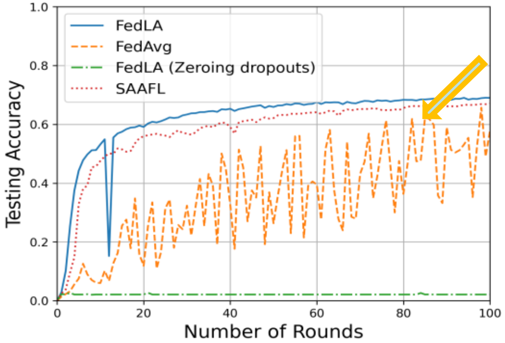


$k_r = 0.8, f = 0.2$



$k_r = 0.9, f = 0.3$

Accuracy over EMNIST dataset



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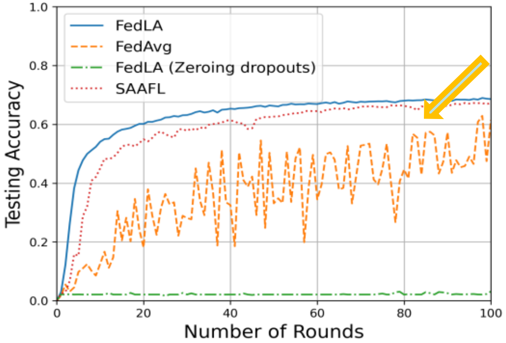

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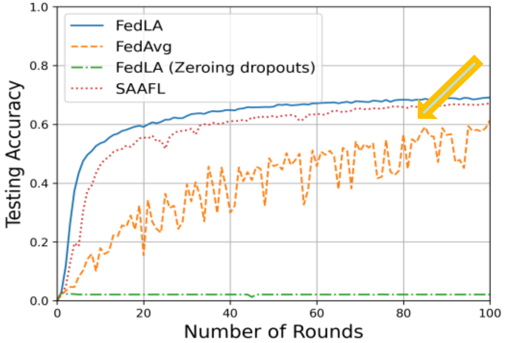
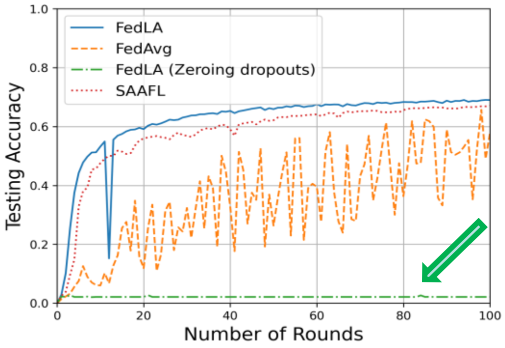


$k_r = 0.9, f = 0.3$

- **FedAvg** gets unstable with higher non-IID distribution

Accuracy over EMNIST dataset



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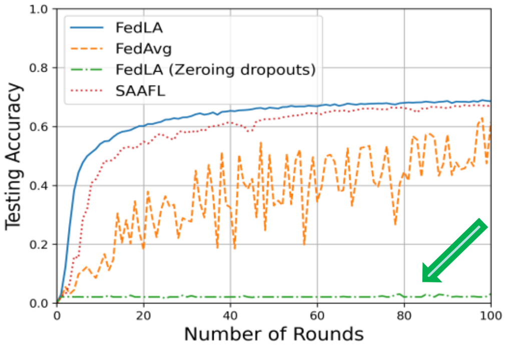
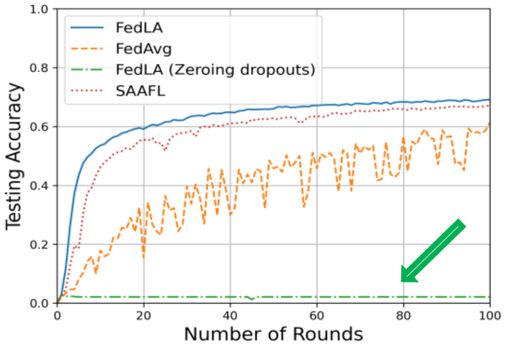


$k_r = 0.8, f = 0.2$



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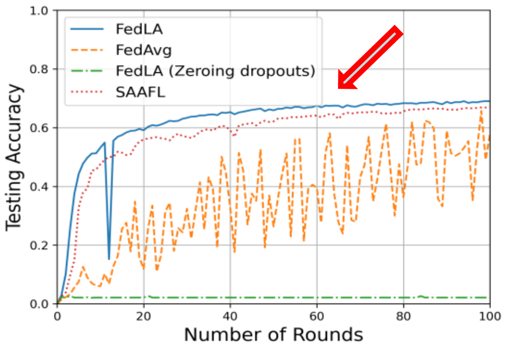

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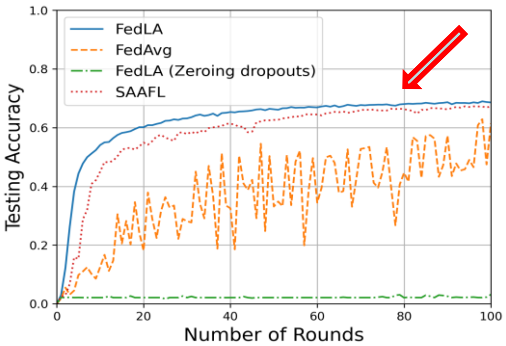

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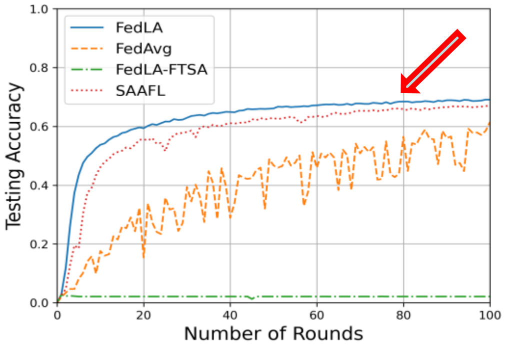

$k_r = 0.9, f = 0.3$

- **FedAvg** gets unstable with higher non-IID distribution
- **FedLA zeroing dropouts** performs worst
- **SAAFL** accuracy ≈ **FedLA** accuracy

SAAFL overhead

Computation and communication cost of **SAAFL** with
$\boldsymbol{n = 100, k_r = 0.9, f = 0.3}$

SAAFL	Server $(\boldsymbol{ms})$	Client $(\boldsymbol{ms})$	BW $(\boldsymbol{KB})$
*Offline-phase	939	287	28.86
FL Round	46879	6454	25.15

* Consists of Key Setup and $\boldsymbol{w_{FedLA}}$ calculation phases

Conclusion & Future Work

- **Conclusion**
  - **FedAvg** for IID data, **FedLA** for non-IID data
  - **SAAFL**: **SA** solution for **FedLA**
    - SA for model and label privacy
    - Dedicated fault-tolerance scheme for dropouts
    - Handles non-selected clients as dropouts
  - **FedLA** performs better than **FedAvg** for non-IID data
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- **Future Work**
  - Addressing stronger threat model: malicious clients or aggregator

Thank you

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